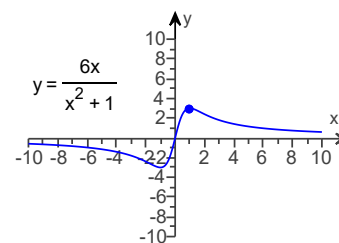


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Course: Calculus Beta

Assignment: MyLab Math - Uge 1

Find the tangent lines to Newton's serpentine (graphed here) at the origin and the point (1,3).



To find the tangent line to the curve $y = \frac{6x}{x^2 + 1}$ at the origin, first find the slope of the curve.

To find the slope of the curve, use the Derivative Quotient Rule, which states if u and v are differentiable at x and if $v(x) \neq 0$, then the quotient

$$\frac{u}{v} \text{ is differentiable at } x, \text{ and } \frac{d}{dx} \left(\frac{u}{v} \right) = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}.$$

Apply the Derivative Quotient Rule.

$$\frac{d}{dx} \left(\frac{6x}{x^2 + 1} \right) = \frac{(x^2 + 1) \frac{d}{dx}(6x) - (6x) \frac{d}{dx}(x^2 + 1)}{(x^2 + 1)^2}$$

To find $\frac{d}{dx}(6x)$, use the Constant Multiple Rule and the Power Rule.

The Constant Multiple Rule states if u is a differentiable function of x , and c is a constant, then $\frac{d}{dx}(cu) = c \frac{du}{dx}$.

The Power Rule states if n is any real number, then $\frac{d}{dx}x^n = nx^{n-1}$.

Find $\frac{d}{dx}(6x)$.

$$\frac{d}{dx}(6x) = 6 \frac{d}{dx}(x) = 6$$

To find $\frac{d}{dx}(x^2 + 1)$, use the Derivative Sum Rule, which states if u and v are differentiable functions of x , then their sum $u + v$ is differentiable

at every point where u and v are both differentiable. At such points, $\frac{d}{dx}(u + v) = \frac{du}{dx} + \frac{dv}{dx}$.

Apply the Derivative Sum Rule.

$$\frac{d}{dx}(x^2 + 1) = \frac{d}{dx}(x^2) + \frac{d}{dx}(1)$$

To find $\frac{d}{dx}(x^2)$, use the Power Rule, which states if n is any real number, then $\frac{d}{dx}x^n = nx^{n-1}$.

Find $\frac{d}{dx}(x^2)$.

$$\frac{d}{dx}(x^2) = 2x$$

To find $\frac{d}{dx}(1)$, use the Derivative of a Constant Function Rule, which states if f has the constant value $f(x) = c$, then $\frac{df}{dx} = \frac{d}{dx}(c) = 0$.

Find $\frac{d}{dx}(1)$.

$$\frac{d}{dx}(1) = 0$$

$$\text{Thus, } \frac{d}{dx}(x^2 + 1) = 2x.$$

$$\text{Use } \frac{d}{dx}(6x) = 6 \text{ and } \frac{d}{dx}(x^2 + 1) = 2x \text{ to find } \frac{dy}{dx}.$$

$$\begin{aligned} \frac{dy}{dx} &= \frac{(x^2 + 1) \frac{d}{dx}(6x) - (6x) \frac{d}{dx}(x^2 + 1)}{(x^2 + 1)^2} \\ &= \frac{(x^2 + 1)(6) - (6x)(2x)}{(x^2 + 1)^2} \\ &= \frac{6 - 6x^2}{(x^2 + 1)^2} \end{aligned}$$

Now, find the slope at $x = 0$ and $x = 1$.

To find the slope at $x = 0$, substitute 0 for x in the derivative.

$$\begin{aligned} \left. \frac{dy}{dx} \right|_{x=0} &= \left[\frac{6 - 6x^2}{(x^2 + 1)^2} \right]_{x=0} \\ &= \frac{6 - 6 \cdot 0^2}{(0^2 + 1)^2} \\ &= 6 \end{aligned}$$

To find the slope at $x = 1$, substitute 1 for x in the derivative.

$$\begin{aligned} \left. \frac{dy}{dx} \right|_{x=1} &= \left[\frac{6 - 6x^2}{(x^2 + 1)^2} \right]_{x=1} \\ &= \frac{6 - 6 \cdot 1^2}{((1)^2 + 1)^2} \\ &= 0 \end{aligned}$$

Now, find the equations of the lines through the origin and (1,3).

Find the line through the origin with slope $m = 6$.

$$\begin{aligned} y - 0 &= 6(x - 0) \\ y &= 6x \end{aligned}$$

Find the line through (1,3) with slope $m = 0$.

$$\begin{aligned} y - 3 &= 0(x - 1) \\ y &= 3 \end{aligned}$$

The equation of the tangent line at the origin is $y = 6x$, and the equation of the tangent line at the point (1,3) is $y = 3$.